

Military personnel

with STEPHAN RUDZKI, TONY DELANEY and ERIN MACRI

*The limbs of soldiers are in as much danger from the ardor of young surgeons
as from the missiles of the enemy.*

Confederate surgeon Julian John Chisholm's observations on the limitations of medical care for military personnel in the American Civil War (1861–1865)

In the United States (US) Army, injuries and injury-related musculoskeletal conditions are responsible for more than 660 000 visits annually.¹ Injuries are the biggest cause of disability in the US Army.¹

In this chapter, we discuss the principles of sport and exercise medicine that will help you provide services to a military population.

We introduce:

- the special culture of the military environment
- epidemiology of military injuries
- common military injuries with a sport and exercise medicine focus (i.e. not major trauma).

SPECIAL MILITARY CULTURE

Clinicians who provide medical and physiotherapy care in a military setting have a challenging task. An important difference between civilian and military practice is the compulsory nature of physical training. All military recruits undergo exercise regimes that are designed not only to improve their fitness, but also to prepare them physically and psychologically for extreme environments, discomfort and pain (Fig. 45.1). Military trainees have not always had a background of physical training. Increasingly, young adults come into the military with little experience of contact sports or regular physical activity. It is challenging to bring them to a required level of strength and aerobic fitness without causing injury.

The traditional military culture generally encouraged personnel to continue exercising despite early warning signs of pain that would cause civilian athletes to stop or slow down. Many recruits conceal injuries until graduation for fear of medical 'back squadding' (recruits held back

in training) and derision from fellow recruits and their superiors. A soldier who finishes a forced march by walking on a broken ankle for 20 km with a fully weighted rucksack receives accolades and respect from peers and superiors.

Further to the drive towards pushing through pain, soldiers with a history of previous injury demonstrate an increased pain threshold—they feel less pain for a given stimulus compared to controls.^{2,3} Pain thresholds may also be related to environmental and psychological factors



Figure 45.1 Physical training in the military takes place within a distinct culture and environment
ISTOCKPHOTO

(e.g. the common experience of World War II soldiers reporting no pain during the heat of battle despite severe injuries).⁴

As a consequence of these changes in experiencing or reporting of pain, military members often present to clinics with musculoskeletal injuries that are severe and debilitating, requiring longer periods of rehabilitation. Patients may present stoically, making it challenging to determine the true severity of pain or injury.

Because of the cultural overlay and late presentation in at least some military settings, it is common for patients to present with multiple concurrent pathologies. It is therefore very important to take a thorough history and physical examination (Chapter 14 and 15) on the initial presentation even if under time pressures. Concurrent pathology should always be considered in a patient whose progress is slow or who does not respond to treatment considered appropriate for the initial diagnosis.

EPIDEMIOLOGY OF MILITARY INJURIES

Recruit populations have an especially high incidence of injury and attrition. In New Zealand, the injury rate for recruits was more than five times that of trained personnel.⁵ High attrition rates represent a significant cost. An injured soldier cannot perform his or her duties even if physically fit, and a moderate level of injury can impinge on the combat readiness of individual units.

US military studies have shown that training injury incidence rates range from 120 to 144 injuries/100 soldiers/year⁶ in infantry, special forces and ranger units and 360/100/year for Naval Special Warfare training.⁷

These rates are far greater than the rates of injury in wars themselves. The Watson Institute monitors the cost of war: more than 970 000 Iraq and Afghanistan veteran disability claims were registered with the US Department of Veterans Affairs as of April 2016.⁸ Rates of musculoskeletal injury in combat are estimated at around 3 per 1000 deployed soldiers per year. However, when a group of soldiers is monitored prospectively the rate is 10 times as high.

One of the larger reports of the type of musculoskeletal injuries highlights the proportion of fractures when soldiers are in the combat setting (as distinct from the training setting when soft tissue injuries dominate) (Fig 45.2).⁹ The most common sites of fracture are humerus, femur, tibia and fibula.

Injuries are not just a problem within a barracks setting. Non-battle injuries in US service personnel have been a major cause of medical evacuation from Iraq and Afghanistan. Referred to as DNBI (disease and non-battle injury), these conditions had an incidence rate of 130 per 1000 combat years and resulted in over 80% of the evacuations required from Afghanistan and Iraq in 2001–2012.^{9,11} The importance of rehabilitation is underscored

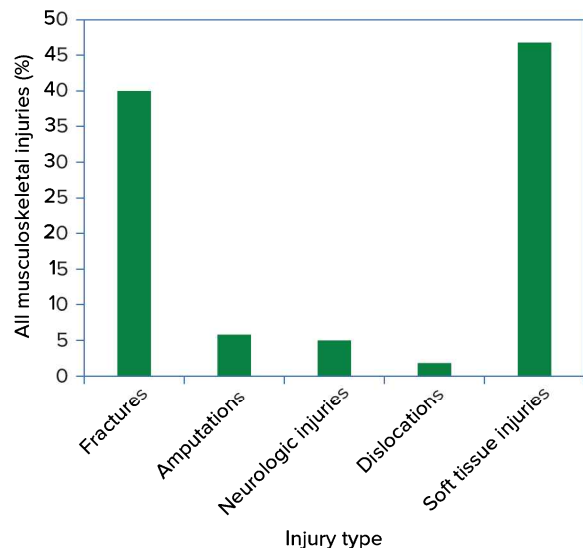


Figure 45.2 Fractures and soft tissue injuries provide the bulk of military injuries in the combat setting¹⁰
ADAPTED FROM BELMONT ET AL.⁹

by the fact that >90% of these injured military personnel returned to duty. Also relevant to the sports medicine community is that 50 000 anterior cruciate ligament (ACL) injuries were sustained among almost 2 million US troops during 10 years in Afghanistan and Iraq.⁹

Historically, the lower limb comprised the bulk of combat injuries (Table 45.1a), but recent wars have seen an increase in spinal/trunk injuries and spinal cord injury as a result of more powerful explosives.^{9,17} The back, knee and wrist/hand are body regions commonly affected in those medically evacuated. This pattern of military injuries (i.e. being primarily lower limb in nature) presents in contrast to the Australian civilian workplace where back injury is the most common injury (25%) followed by the hand (14.3%) and finally lower limb injuries (10.8%).¹⁸ Overuse is the most common cause of military training injuries.^{6,7,14,15}

Wearing combat body armour is a risk factor in the development of low back pain during a military deployment.^{19,20} In a prospective cohort study of US troops who deployed to Afghanistan for 12 months, moderate or worse low back pain occurred in 22% (175/805) and there was some back pain in 77% of the soldiers. Risk of back pain was sharply increased (relative risk >4.0) if the combatant wore body armour for greater than 6 hours a day or wore equipment weighing >32 kg. The incidence of low back pain in deployed US troops increased 20 times from 1.5/1000 in 2001 to 29.8/1000 in 2010. See Chapter 29 for discussion of the biopsychosocial factors that contribute to back pain.

Table 45.1 (a) Types and incidence of injuries by country

Body part	US infantry ¹²	US military ¹³	Australian recruits ¹⁴	Australian Army ¹⁴	South African recruits ¹⁵	New Zealand recruits ⁴
Ankle/foot	11.6%/20.8%	13%	18.3%/11.9%			35%
Knee/lower leg	17.6%/15.1%	22%	32.1%/7.3%			16%
Low back	10.2%	20%				
Spine				15.2%		
Lower limb				39.6%	80%	
Upper limb				19.4%		

(b) Cause of injuries by country

Cause	US infantry ¹²	Afghanistan/Iraq medical evacuations ¹³	Australian recruits ¹⁴	Norwegian recruits ¹⁶	New Zealand recruits ⁴
Training-related	47%	19–21%	19.2%		
Falls/jumps		18%			
Motor vehicle-related		12–16%			
Running			36.6%		28%
Obstacle course			14.6%		
Basic training				20–25%	
Acute overexertion					37%
Team sports					25%

Injuries cause disproportionate morbidity in young military populations. In a US military population, injuries accounted for 56% of sick-call diagnoses, but caused nearly 10 times the number of limited-duty days as illness. Soldiers with lower extremity running injuries spent seven times more days on a restricted duty profile than those with non-running injuries.²¹ In outpatient clinics, 80–90% of all limited-duty days accrued by US Army trainees and infantry soldiers were the result of training-related injuries (Table 44.1b).⁶

COMMON MILITARY INJURIES

The range of injury and illness facing clinicians who serve the military greatly exceeds that usually seen in civilian medicine. In the military, individuals are exposed to extremes in temperature, biological and chemical agents, and communicable diseases (e.g. sexually transmitted, tropical). Traumatic injuries can result in concussion or traumatic brain injury, spinal cord injury, limb amputation, and injuries that can affect multiple systems and often require urgent medical care. For example, significant risk for injury occurs in diving operations, including

submariner evacuation and rescue. However, the majority of the injuries tend to be related to environmental exposure (e.g. barotrauma, decompression illness, cold exposure, marine life exposure) rather than to mechanical mechanisms of injury, and thus most diving injuries are of a non-musculoskeletal nature.^{22, 23}

Psychosocial factors can influence the recovery trajectory of military sports injuries (see also Chapter 5). Experiencing deployment can affect emotional and psychological health, even in situations of peacekeeping missions, resulting in issues ranging from fatigue and exhaustion to depression, post-traumatic stress disorder and suicidal ideation. Mental health is a major issue in military medicine and suicide rates are higher among military populations than among civilians.⁹

The following discussion will address musculoskeletal and sports injuries. However, maintain a broad perspective when assessing sports injuries as often these other factors present at the same time. Many 'injury' chapters (Part B) in this volume are relevant to the military population. This chapter highlights common specific military issues and treatment that is particular to this population.



Figure 45.3 Running is a predominant source of exercise in many forces. Competition is high and environments are usually extreme compared with recreational running programs ISTOCKPHOTO

Overuse injuries of the lower limb

There are three peaks of overuse lower limb injuries in the military. The first and greatest is among recruits, the second in trained soldiers preparing for special forces selection (Fig 45.3) and the third is older soldiers training to pass annual fitness assessments.

The most common lower limb overuse injury is leg pain due to medial tibial stress syndrome (see Chapters 4 and 38). Early identification with correction of training errors and biomechanical factors often leads to rapid resolution. Recalcitrant medial tibial stress syndrome may require customised orthoses, gait retraining and more appropriate footwear where possible.

A significant proportion of recalcitrant medial tibial stress syndrome may be due to underlying exertional compartment syndrome (see Chapter 38). We have encountered resting compartment pressures >50 mmHg (normal resting <15 mmHg) in military trainees with largely medial tibial pain and tenderness. Changes in running technique (shortening stride length, landing more towards the forefoot) decreased leg pain and increased pain-free running distance in US Army recruits

at the United States Military Academy at West Point (see Chapter 38).²⁴⁻²⁶

The incidence of stress fracture in military recruits varies from 1-30%, with female rates generally higher than male rates.^{15, 27-32} The most common sites of stress fracture are the tibia, metatarsals, pelvis and femur.³³ Previous stress fracture and lower baseline levels of weekly exercise are well-recognised risk factors.³² There may be genotypic predisposition.³²

To prevent stress fractures, as well as other injuries, aim to have recruits enter physical training with a reasonable level of fitness (as measured by the 20 m shuttle run). The issue of fitness at entry and injuries is discussed in more detail later in this chapter. Correction of biomechanical faults, and reduction of training to 60-70% of the injury-causing load, will contribute to rapid recovery and return to full training. Complete rest is contraindicated as this leads to loss of cardiorespiratory fitness and often depression. Training should change from large marching and running volumes to cycling, rowing and swimming pool work. For more on stress fracture aetiology see Chapter 4 and for treating specific stress fractures, please see the index. Medial tibial stress syndrome, exertional compartment syndromes and bone stress injuries can coexist. All personnel presenting with lower limb injuries should be evaluated for referred pain from the spine (Chapters 5, 6 and 29).

Blister injuries

Blister injury from endurance marching has been a major problem for infantry soldiers throughout history. Among Canadian military, there was a 43% incidence of blisters during a 56 km night march carrying an 11 kg load.^{34, 35} High rates of blister injury occur in recruits in the US Marine Corps and US Navy.³⁶ The morbidity associated with blisters should not be underestimated.³⁷

Blisters are caused by shear forces acting on the skin, mainly due to friction while braking. This causes a mechanical split in the skin layers, which subsequently fills with fluid. Moist (sweaty) skin is most prone to blister formation because of increased friction and softening of the hard outer layer of skin (stratum corneum).³⁸ Braking forces on the foot increase with increased rucksack load.³⁹

Sock type reduces the incidence of blisters. During a 3-day road march in the United Kingdom (UK), soldiers wearing a nylon inner sock had fewer blisters per person than those with either a single or double woollen sock.⁴⁰ A study in runners suggests that an acrylic sock results in fewer and smaller blisters than a cotton sock.³⁸ A joint US military task force recommended the use of synthetic-blend socks to prevent blisters.⁴¹

In addition to type of sock, use of an inner sock can reduce the severity and associated morbidity of blister

JOURNAL OF THE ROYAL ARMY MEDICAL CORPS

An International Journal of Military Medicine



Fig 45.4 The *Journal of the Royal Army Medical Corps* (cover shown) along with *Military Medicine* provide valuable research and education

A 2016 COVER IMAGE, COURTESY OF THE JOURNAL OF THE ROYAL ARMY MEDICAL CORPS

injuries compared to a single sock.⁴² Foot powder does little to reduce blister incidence in recruits and may even increase the incidence of blister formation.⁴³

Boots have traditionally been blamed for the problem of blisters, but changes in boot fit, style and composition have had little effect on blister incidence.^{40, 44, 45} Nevertheless, the following strategies are recommended for boot fitting and blister prevention:⁴²

- wear a sock, sock liner and insoles (if applicable)
- have a load on your back when fitting boots
- measure both feet; length and width
- fit length first
- push the foot forwards in unlaced boot; ensure 1.5 cm space behind the heel
- push the foot backwards; ensure 1.5 cm from the longest toe to the end of boot
- ensure the boot width over the arch is snug but not tight
- fit the boots at the end of a day's activity, when the foot has spread
- apply sports tape directly to the feet before a long march; this may prevent blisters

- use off-the-shelf heat-mouldable orthoses to reduce shearing loads to the sole of the foot
- reduce heel lift and blistering by tightening boot laces over the dorsum of the foot and tying a reef knot to lock the lace at the anterior ankle joint.

Parachuting injuries

Military parachuting has the potential to cause severe injuries.⁴⁶ Military parachutists use static line parachutes with limited manoeuvrability. In conditions where the prevailing winds exceed 5–8 knots, the parachutist has limited ability to reduce speed on landing. Dirt landing strips and airports are hazardous landing areas. Night descents are associated with a greater rate and severity of injury.⁴⁶ Paratrooper body weight influences risk of injury, particularly when the drop zone is on land.⁴⁷ There appears to be a relatively high rate of thoracic spine injuries in military parachutists jumping with operational loads.

Military round parachutes should not be used in wind conditions over 10 knots for trainees and 13 knots for trained static line paratroopers. If the wind on a drop zone is a steady 10 knots, there is a high probability of 15–18 knot gusts every 10–15 minutes. A 20 knot wind carries a parachutist into an obstacle at four times the force of a 10 knot wind (i.e. $F = 1/2mv^2$). There is often real or perceived pressure to complete training or military exercise jumps in marginal wind conditions. Most military mass parachuting injuries occur when a strong wind gust comes through after the jump has been initiated. Our experience is that the cost of aborting a jump due to wind is dwarfed by the financial and human cost of injuring highly trained paratroopers or special forces operators, many of whom will not return to pre-injury duties.

PRACTICE PEARL

Prophylactic ankle bracing significantly reduced ankle sprains and fractures in US Army paratroopers.⁴⁸

Prophylactic ankle bracing significantly reduced ankle sprains and fractures in US Army paratroopers⁴⁸ and it also reduced the amount of time spent on medically restricted duties.⁴⁹ It has been estimated that for every dollar expended on a parachute ankle brace, a saving of between US\$7–9 could be achieved in medical and personnel costs.⁵⁰

The ageing defence forces

Retirement ages are extending to 60 for full-time defence personnel and up to 65 for some reservists. With ageing, incidence of injury may increase as may risk of

central obesity, insulin resistance, hyperlipidaemia, hypertension, cardiovascular disease and neoplasia. Attention to lifestyle, exercise and diet is equally as important as in the general population.

INJURY PREVENTION IN THE MILITARY

This chapter extends the principles of injury prevention outlined in Chapter 12 and in specific injury chapters (e.g. Chapter 35, prevention of ACL injuries) to focus on prevention in the military setting. Risk factors for injury can be divided into intrinsic (specific to an individual, e.g. age, height, foot shape, sex and body weight) or extrinsic (training errors, footwear or environmental considerations such as weather or running surface). This distinction is important in a military context because military forces must recruit from a civilian pool with a wide range of physical and athletic abilities, and seek to train these new members to achieve and maintain a standard of fitness relevant to the physical demands of the profession. There is little or no scope to select out applicants with intrinsic risk factors, as most military forces are required to take applicants from the broadest possible pool.

During recruit training, very high rates of injury lead to medical discharge and a loss of personnel. Australian recruits who developed an injury during recruit training were 10 times more likely to fail to complete training than those who were not injured.⁵¹ US male recruits who were discharged during training sustained injuries three times more frequently than those not discharged; discharged females sustained injuries at twice the rate of non-discharged females.⁵²

As it is costly to attract and process new recruits, minimising recruit attrition during basic training is an important objective for all military forces. Addressing modifiable extrinsic factors can provide the greatest yield in terms of decreasing injury levels and allowing the broadest range of entrants to successfully enter into a military career.

Injury surveillance

Injury surveillance is the necessary precursor to injury prevention; it allows for the identification of incidence, location, nature and cause of injury. Population-based data are necessary in order to target interventions appropriately.

Ideally, military services should adopt a public health approach to injury prevention as a framework.⁵³ The public health approach is characterised by four steps:

1. surveillance
2. risk factor identification
3. intervention implementation
4. evaluation (Fig. 45.5).⁵⁴

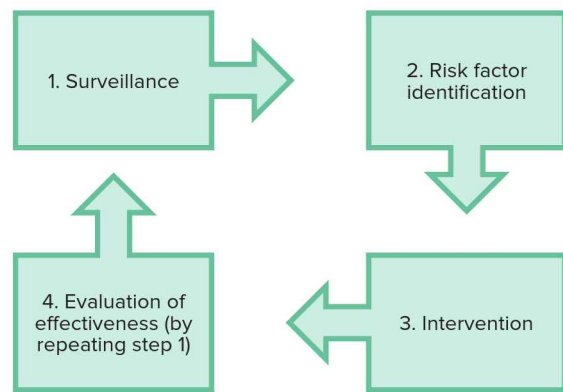


Figure 45.5 The public health model that is appropriate for the military setting. Note the similarities with the injury prevention model outlined in Chapter 12

Surveillance data are fundamental to priority setting, but are dependent on reliable and valid data collection and dissemination. Without such data, it is nearly impossible to measure the impact of interventions.⁵³

The aim of any military injury surveillance system should be to provide commanders with accurate and reliable information on the type, nature and cause of injury, in conjunction with well thought out strategies for injury prevention. The army dictum 'do not come to me with problems, just solutions' is especially pertinent in military settings.

The public health approach to injury prevention was utilised to identify and eliminate ACL ligament ruptures at an Australian Army Recruit Training Centre as detailed below.^{55, 56}

1. *Surveillance*: data identified a greater than expected number of ACL ruptures occurring in an obstacle course landing area.
2. *Risk factor analysis*: rubber matting installed to provide shock absorption was causing ACL injuries due to increased friction and 'stickiness' on landing.
3. *Intervention*: removal of the rubber matting and its replacement with river pebbles.
4. *Evaluation*: no further ACL ruptures at this particular location after removal of the rubber matting.

The success of this approach highlights the utility of following a public health approach. A similar approach has been adopted with other key risk factors including initial level of aerobic fitness and running distance (see relevant sections below).

Despite evidence to support strategies to prevent injuries in a military setting, effecting change is challenging as in all walks of life.⁵⁷ Officers and non-commissioned officers (NCOs) may seek to maintain and

honour traditions, and may resist changes in training regimes or minimum fitness standards. For example, there are conflicting opinions within the military as to whether fitness standards should be different for men and women if they are working in the same trade. One argument is that a combat engineer needs to meet the standards for being a combat engineer regardless of gender.

To summarise, the implementation of changes must consider the cultural environment of the military and a strategic plan implemented in order to ensure a smooth and effective transition.⁵⁸ In recent times, Western militaries have moved towards gender-free assessments of physical capability that are valid predictors of work performance. Australia has removed all sex barriers to military employment in 2016. All jobs in the Australian Defence Force are open to women who are able to pass the relevant occupation-specific tests. The effects this change will have on injury type, frequency and severity will be the subject of prospective study and evaluation.

Sex as a risk factor for injury

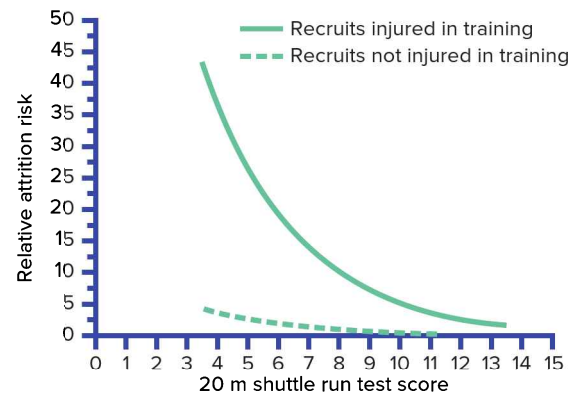
In Australian, British and American troops, there are higher rates of injury and illness among female soldiers than their male counterparts.⁵⁹⁻⁶² Overall morbidity of injury in women is also greater.⁶³ Medical discharge rates are higher among women in both Australia and Britain.^{60, 61}

An increased risk of injury was first identified in 1976 at the US Military Academy at West Point with the introduction of female cadets.⁶⁴ On average women had lower aerobic fitness than men (VO_2max , 46 mL/kg/min versus 59 mL/kg/min) and this variable was associated with injury incidence.⁶⁵

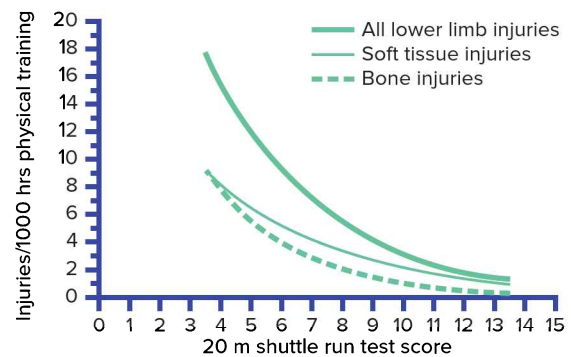
When stratified by 1 mile run time, the risk of injury between men and women was comparable, with the highest risk of injury in the slowest group.⁶³ Of note was that 51% of the women were in the group with the longest run time compared to 1% of the men. Lower peak VO_2 is also an independent risk factor for time-loss injury.⁶⁶ For a standardised level of physical activity, UK male recruits worked at a lower cardiovascular strain than females (24% vs 33% of heart rate reserve), and this additional cardiovascular strain is believed to increase fatigue, thus predisposing to injury.⁶² Sex is therefore not an independent risk factor for injury when controlled for aerobic fitness levels among recruits.⁶¹

Aerobic fitness

The initial level of aerobic fitness of a recruit has been shown to be a predictor of both injury and successful completion. In an Australian Army recruit cohort, aerobic fitness was measured using the 20 m shuttle run test (SRT).^{67, 68} Lower scores on the 20 m SRT were related to



(a)



(b)

Figure 45.6 (a) The risk of recruit attrition based on 20 m SRT (b) The rate of recruit injury based on 20 m SRT
REPRODUCED FROM POPE⁷⁰ WITH PERMISSION

increased risk of attrition (Fig. 45.6a) as well as increased risk of injury (Fig. 45.6b).⁵¹ The least fit soldiers (20 m SRT score 3.5) were 20 times more likely to not complete training than the fittest recruits (score 13.5). If injured, fit subjects were 25 times more likely to recover from their injury than the least fit subjects with an injury. Similar relationships between fitness, injury rates and attrition have been seen in British Navy and Army recruits.^{61, 69}

One and 2 mile run times (both proxies for aerobic fitness) are significant predictors of injury in US Army and infantry recruits, respectively.^{12, 63, 71} As mentioned above, recruits of equivalent aerobic capacity have similar rates of injury. However, in a female cohort of US Marine Corps recruits, a slower 2 mile run time was associated with increased incidence of stress fractures. It was suggested that stress fractures may be reduced if women participated in pre-training activities designed to improve aerobic fitness.³³

It should not be surprising that faster runners get fewer injuries. They are, almost by definition, good runners, and therefore less likely to sustain a running-related injury. Fitter individuals also do not experience

the same level of physiological strain during recruit training as those with lower levels of aerobic fitness. Less-fit individuals work closer to their maximum workload during recruit training and thus are at increased risk of injury.

The situation of a variable aerobic fitness start point and fixed endpoint could be viewed as a recipe for injury.^{51,63} Ensuring that recruits arrive at training centres with a minimum level of aerobic fitness is a logical strategy to reduce the risk of injury and subsequent attrition. A screening 20 m SRT test was introduced at all Australian Army recruiting centres in 1997, and only recruits who scored higher than level 7.5 on the 20 m SRT were allowed to proceed to the recruit training battalion. The Royal Navy used a 2.4 km treadmill run to assess initial aerobic fitness.⁷² Following introduction of the screening test, the pass rate in recruit training increased by 10%, and the number of recruits applying for voluntary release decreased from 15% to 6%. Overall, pre-enlistment aerobic testing positively impacted on recruit training pass rates.

The US Army tried a slightly different approach, in which they identified recruits with poor aerobic fitness at entry and then provided them with an in-service remedial physical fitness program before starting basic combat training (BCT). Women who undertook this remedial training then began BCT with similar mean 2 mile run times compared with non-remedial women, and had similar graduation success and time-loss injury rates.

Men with poor aerobic fitness (slower 2 mile run time) also undertook remedial training, but they did not improve their fitness to the level of normal entry male recruits. These men began BCT with considerably slower mean 2 mile run times than normal entry male recruits, and were ultimately less likely to graduate and more likely to suffer a time-loss injury.⁵² Thus, the remedial program improved initial fitness levels in female, but not male, recruits within a predetermined time period.

These data suggest to us that encouraging potential recruits to achieve a minimum level of aerobic fitness before presenting to a recruit training centre may result in better outcomes than investing in remedial training once enlisted. Regardless of the method employed, overall evidence suggests that improving aerobic fitness levels prior to basic training could help to decrease the risk for injury and attrition.³³

Body composition

If aerobic fitness level influences risk of injury and subsequent attrition rates in the military, would body composition also be a contributing factor? It has been noted that, in the US Army's view, 'obesity is associated with being unfit and unsoldierly'.⁶³ This is a view shared

by many military forces, and most have programs in place to assist members to maintain their weight within prescribed ranges of body mass index (BMI) or percentage body fat.

The US Army weight control program has the stated aims of ensuring that soldiers are physically fit for their combat mission and that they present 'a trim military appearance'. US Army regulations consider body composition to be a subcomponent of fitness.⁶³ Using defined body fat standards, it has been estimated that 5% of eligible men and 30% of eligible women would be excluded from enlistment in the US military based on failing to meet these standards.

There is a clear relationship between body composition and performance on run tests: 1 and 2 mile run times increase with increasing percentage body fat.⁶³ For males, the highest percentage body fat group had more injuries; however, in females, the leanest had more injuries.⁶³

Current cohorts of young adults are less fit than previous generations of young adults entering military service in the US, as mean times for 1 and 2-mile runs have increased since 1987.⁷³ US Army recruit height, weight and BMI values have progressively increased, with increases of both body fat and fat-free mass in male recruits. Body composition data on female recruits does not show a consistent trend.⁷³

In Norwegian recruits, BMI was an independent risk factor for injury, while height and weight were not associated with injury.¹⁶ In Australian recruits, none of these measures predicted injury.⁵¹ Also, despite the US Army's views on obesity, there were no significant differences in attrition between Army recruits who met body weight standards and those who did not, regardless of sex, provided they met physical fitness standards on entry.⁷⁴

PRACTICE PEARL

In considering the conflicting evidence available regarding BMI as an independent risk factor for injury, it is important to recognise that efforts might best be served focusing on overall fitness levels rather than body composition per se in assessing suitability for entry into the forces.

Previous injury

Many studies have linked history of injury and likelihood of a new injury. After adjusting for weekly distance, there was a 65% increased risk of injury in runners with a history of previous injury⁷⁵ and a doubling of the risk in those with previous injury.⁷⁶ After adjusting for age and sex, previous injury was a strong predictor of injury

with an odds ratio of 1.5.⁷⁷ In one study of runners training for a marathon, half of the participants who reported an injury had sustained a previous injury and 42% had not completely rehabilitated before starting a training program.^{78, 79} Controlled rehabilitation reduced the number of injuries in soccer.⁸⁰

Previous injury and incomplete rehabilitation are therefore strong risk factors for recurrent injury. Rehabilitation is complete when an athlete is free from pain, muscle strength has returned to the pre-injury level and joint range of motion is restored.⁸⁰

Clinicians responsible for the care of soldiers have a responsibility to ensure that complete recovery from injury has occurred before clearing them to return to full duties. Sports medicine clinicians appreciate that musculoskeletal injuries can only recover fully with adequate functional training.

PRACTICE PEARL

Consistent with the principle of specificity, a soldier will only recover sufficiently to return to full duties if given a functional training regime to prepare for full duties.⁸¹

Weekly running distance

Overtraining, and in particular, excessive running distance, is the most significant underlying cause of lower limb injuries in athletic populations.^{82, 83} The only risk factor for injury consistently identified in running studies is weekly running distance. There is an almost linear relationship between increasing weekly distance and the incidence of injury in both men and women.^{75, 76, 84-87} A sudden increase in weekly distance without a gradual build-up (or 'spike' of >10% per week) is considered a training error.^{87, 88} Training errors such as running too frequently, too fast or for too long are major causes of injury in both beginner and experienced runners, accounting for up to 60% of injuries.^{82, 87, 89, 90} We recommend Dr Tim Gabbett's review of sports injuries and training loads to all who work in the military setting (Chapter 12).⁹¹

Reducing weekly run distance

Most military forces use running tests ranging from 1.6 km (1 mile) to 5 km (3 miles) to assess physical fitness. The distance of the Australian Army physical training test has been reduced from 5 km to 2.4 km on the basis that reducing the test distance might reduce the training distances and result in a concomitant reduction in overtraining. However, attempts such as these have met with varying degrees of success worldwide.



Figure 45.7 Marching has historically been a mainstay of military forces, both in training and in active deployments
ISTOCKPHOTO

In US studies spanning the Marine Corps, Army and Navy, recruits given lower mileage running programs sustained fewer injuries with no subsequent differences in performance on run tests or other physical fitness tests.⁹²⁻⁹⁵ One study demonstrated a 54% reduction in stress fracture incidence compared to the high-mileage group.⁹² The US Department of Defense estimated in 1995 that the reduction in running mileage had saved US\$4.5 million in medical care costs and nearly 15 000 training days annually by preventing stress fractures alone.⁹⁶

Cross-training and other program modifications

There are effective alternatives to running to condition recruits. Marching with rucksacks in place of running was investigated in Australia with subsequent reduction in injury rates and morbidity, while improvements in performance on run tests and VO_2max were as good as a group given a traditional running program (Fig. 45.7).^{14, 97} Load bearing or pack marching is more occupation-specific to the military environment than distance running in a singlet, shorts and shoes. There is considerable gain in aerobic conditioning from shorter, high-intensity training. Given the variety of careers in the military, with varying physical and intellectual demands, imposing a single rigid training regime on all recruits, regardless of physique, sex, age and aspiration, is unrealistic.

Some level of running is essential in military training programs, but the design of running programs may reduce the risk of injury. Interval training and periodisation are other training modifications that have been successfully used to reduce injuries.^{60, 94, 98}

In Australia, sport and exercise physician Dr Stephan Rudzki published the results of an intervention that consisted of:

- discontinuing road runs as formed groups
- introducing 400–800 m interval training
- reducing test run distance from 5 km to 2.4 km
- standardising route march speed, and
- introducing deep water running as an alternate aerobic activity.

The incidence of injuries decreased 46% among men but increased in women. Overall, there was a striking cost saving because of the much lower rates of medical discharge and medical investigation.⁶⁰

In a US study, recruits were divided into ability groups based on initial 2 mile run times. Running speed was managed to keep effort at between 70–83% of VO_2max . This modified program was compared with a traditional training program. The proportion of recruits who failed the Army Physical Fitness Test (APFT) was lower in the modified program than the traditional training group (1.7% vs 3.3%). After adjustment for initial fitness levels, age and body mass index, the relative risk of an injury in the traditional training group was about 50% greater in both men and women.⁹⁹

Training modifications nearly abolish the incidence of pelvic stress fractures in female recruits.⁹⁸ Interventions included a reduction in route march speed from 7.5 km/h to 5 km/h and placing shorter women at the head of the column to reduce stride length. Individual step length was promoted instead of marching in step, march and run formations were more widely spaced, and interval-running training replaced traditional middle-distance runs. The rate of pelvic stress fractures fell from 11% to 0.6%.⁹⁸

Deep water running is an aerobic activity that can rest the lower limbs and periodise impact loading.¹⁰⁰ It is performed in the deep end of a swimming pool, normally with the aid of a flotation vest or belt. This form of training will maintain existing aerobic performance for up to 6 weeks.

To summarise, a comprehensive review of injuries in the US Armed Forces stated that:

Military and civilian research indicates that high running volume significantly increases the risk for lower extremity injury. During initial military training, about 25% of men and about 50% of women incur one or more training-related injuries. About 80% of these injuries are in the lower extremities and are of the overuse type—a condition brought about by training volume overload (generally excessive running relative to initial fitness level and running capability of the individual).⁴¹

PRACTICE PEARL

Reducing running distances among recruits is a logical strategy to reduce injury rates while still attaining minimal physical fitness standards. Ideally, recruits come to training in good condition; this has been associated with fewer injuries.¹⁰¹

Running experience

The risk of running injuries declines with increased running experience.⁷⁵ This appears contradictory in the context of the clear association between running distance and injury, but may be related to experience (i.e. chronic adaptation of tissues to long-term running programs, improved running mechanics with experience) and/or the 'healthy runner' effect.^{82, 90, 102} The 'healthy runner' describes the phenomenon that only those without injury can continue with a running program on a long-term basis. This suggests that 'healthy runners' have the intrinsic characteristics required to become long-term runners in the first place, while others will stop running due to an inability to tolerate the sport. Alternatively, if experience causes reduced injury rates over time, then this would suggest that the reason military recruits tend to have higher injury rates is because they are new to the intensity of the training regime. Either way, runners with less than 3 years of regular running experience have twice the risk of lower limb injury than those with more than 3 years running experience.⁷⁶

Competitive behaviours

Injury incidence is associated with motivation score.¹⁰³ Higher scores correlate with higher incidence of medically treated injury. Runners who compete have higher injury rates than recreational runners, even when adjusting for mileage.^{75, 77} More motivated runners may ignore the first signs of injury.

In military populations, every activity is highly competitive. The US Army places great emphasis on the APFT. Better scores influence prospects for promotion and most US soldiers strive to achieve the maximum score. The Australian Army now utilises a pass/fail result instead of a score, to reduce competitive behaviours and subsequent risk of injury.

Warm-up/stretching

The role of stretching as a component of a pre-activity warm-up, and in preventing injuries in the military, is unclear. As outlined in Chapter 12, in the general sporting population an acute bout of stretching prior to activity results in short-term loss of muscular strength, power and endurance^{104, 105} and does not reduce overall injury risk¹⁰⁶

Military recommendations for injury prevention initiatives

The evidence base for injury prevention recommendations has been reviewed by a joint US military task force.⁴¹ Six interventions had strong enough evidence to be recommended for implementation in the military services and two interventions were not recommended due to evidence of ineffectiveness or harm.

Recommended interventions (sufficient scientific evidence)

1. Prevent overtraining.
2. Perform multiaxial, neuromuscular, proprioceptive and agility training.
3. Wear mouth guards during high-risk activities.
4. Wear semi-rigid ankle braces for high-risk activities.
5. Consume nutrients to restore energy balance within 1 hour following high-intensity activity. (High carbohydrate/protein replacement within 15–30 minutes of finishing high-intensity exercise leads to more rapid muscle and liver glycogen restoration and preservation of muscle mass.)
6. Wear synthetic-blend socks to prevent blisters.

Interventions not recommended (evidence of ineffectiveness or harm)

1. Wear back braces, harnesses or support belts.
2. Take anti-inflammatory medication prior to exercise.

While physical training is necessary to condition soldiers for their occupational tasks, the authors noted that 'in classic military tradition, however, efforts to exceed the standards have contributed to the injury epidemic present today'.⁴¹

In a randomised controlled study in Australian military recruits there was no effect of pre-exercise static stretches on all-injury risk, soft tissue injury risk or bone injury risk.¹⁰⁷ Among Japanese recruits, the total injury rate was the same between those who conducted static stretches and those who did not. However, the incidences of muscle/

tendon injury and low back pain were significantly lower in the stretching group.¹⁰⁸

As it is clear that traditional static stretches do not reduce overall injury rate, warm-ups today should be incorporating light cardiovascular activities such as jogging and dynamic range of motion activities, and progress to sport-specific drills.^{104, 109-111}

PRACTICE PEARL

The wise military physician should never assume that all apparent musculoskeletal pain in the young soldier is due to overuse injury.

Conclusion

Injury is a major problem for most Western military forces, and female recruits bear a disproportionate burden. Although the military population usually is fit and has access to high-quality healthcare, they can still have the systemic, metabolic, infective and neoplastic diseases found in the wider population.

Reducing running distance reduces both the incidence and severity of injury in military recruit populations, without affecting overall physical performance. The entry level of aerobic fitness is a key modifiable risk factor, and screening for physical fitness prior to clearing to participate in basic training programs has reduced both injury and attrition rates. Early identification of injuries and counselling trainees about achievable career paths is an important part of rehabilitation. Not all somatotypes or psyches are suited for infantry, marines or special forces, regardless of pre-recruitment dreams.

Injury prevention strategies should be based on robust surveillance systems using a public health approach to accurately identify risk factors, implement interventions and evaluate outcomes. While the nature of the profession comes with inherent risk, it is possible to reduce injury and improve retention.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

